Candidate Seat no…………

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester B. Tech (ME) Examination

November 2017

ME 305.01/ME 305 Manufacturing Processes-II

Date: 30.11.2017, Thursday **Time: 10.00 a.m. To 01.00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the questions below. (Each carries one mark). **[07]**

(i) Define feed in the facing operation.

(ii) What is the difference between live and dead center?

(iii) What is the function of tailstock?

(iv) What is the function of three jaw chuck?

(v) What is the difference between face plate and drive plate?

(vi) When the half nut or split nut is used in lathe machine?

(vii) What are the major parts of a carriage?

Q - 2. Answer any three questions. **[12]**

(i) What is the need of changing speed on lathe machine? Explain.

(ii) Calculate the time required to machine a workpiece 170 mm long, 60 mm diameter to 165 mm long, 50 mm diameter. The workpiece rotates at 440 rpm, feed is 0.3 mm/rev and maximum depth of cut is 2 mm for turning. Assume total approach and over travel distance as 5 mm for turning operation.

(iii) Explain the importance of operation sequence in Lathe machine with suitable example.

(iv) A hole with 25 mm diameter and 50 mm depth is to be drilled in a mild steel component. The cutting speed can be taken as 40 m/min and feed rate as 0.25 mm/rev. Calculate machining time and material removal rate.

(v) Explain with neat sketch :

1. Counter-Boring
2. Counter-Sinking

Q - 3 Answer any four questions. **[16]**

(i) Sketch & compare up milling and down milling.

(ii) Set the dividing head to mill 16 teeth and 45 teeth on as per the gear blank.

(iii) A slot of 30 mm X 30 mm is to be milled in a workpiece of 300 mm length using a side and face milling cutter of diameter 100 mm, width 30 mm and having 20 teeth. The depth of cut is 5 mm, feed per tooth is 0.1 mm, cutting speed 35 m/min. Calculate the time required for milling.

(iv) Why Quick return mechanism is provided on a shaper? Explain the crank and slotted lever mechanism for the shaper machine.

(v) Explain the grinding wheel – workpiece interaction.

(vi) Sketch a typical broach tooth profile and name its elements.

SECTION – II

Q - 4 Answer the following.

(i) Differentiate drawing out and upsetting operation in forging. **[02]**

(ii) What do you mean by flash in forging? Name the process by which it can be removed. **[01]**

(iii) Arrange the terms in order to have higher to lower cross section areas. Billet, bloom, ingot, slab. **[01]**

(iv) Prove that plastic deformation process is the constant volume manufacturing process. **[03]**

Q - 5 Answer any four questions. **[16]**

(i) Explain steam hammer with neat sketch.

(ii) Show by schematic sketches the process of forward and backward extrusion. Give two examples of components produced by extrusion.

(iii) Prove the equation for maximum draft in rolling process with neat sketch.

(iv) Give any four examples of rolling stand arrangements with figures.

(v) How the metal forming processes differs from other manufacturing processes?

Q - 6 Answer any three questions. **[12]**

(i) A tensile specimen with a 12 mm initial diameter and 50 mm gauge length reaches maximum load at 90 kN and fractures at 70 kN. The minimum diameter at fracture is 10 mm. Determine the engineering stress at maximum load and the true fracture stress. Also find out true strain and engineering strain.

(ii) A 60° bend is required on a sheet metal component. Should the die angle be equal to, more, or less than 60°? Justify your answer with theoretical support.

(iii) Define extrusion ratio. What will be the extrusion ratio during production of 6 mm^2 wire from 200 mm^2 billet. Enlist points which make extrusion process superior than other metal forming processes.

(iv) Determine the die and punch sizes for blanking a circular disc of 20 mm diameter from a C20 steel sheet having thickness 1.5 mm. Also find out the punching force. Allowable shear stress is 294 MPa.

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